

◆ QUICK-COMMERCE PROFITABILITY · AN EMPIRICAL INQUIRY

THE ANATOMY OF *PROFIT LEAKAGE*

A rigorous, end-to-end examination of 947,752 transactions across eight platforms and twelve cities — tracing every rupee of margin from its ledger entry to its ultimate provenance.

SQL → PYTHON → POWER BI

947,752 TRANSACTIONS

21 ATTRIBUTES

8 PLATFORMS

12 CITIES

9.77% ATTRITION

PREPARED FOR THE DISCERNING STAKEHOLDER

Quick-commerce subsists on gossamer-thin margins — whose provenance remained obscure



The velocity-versus-viability tension

Platforms compete on ten-minute delivery, yet both alacrity and discounting exact a cost. Which levers remained genuinely profitable was, until now, indeterminate.



An absence of unified discernment

Revenue, delivery expenditure, discount outlay, and customer sentiment resided in disparate reports — none reconciled against profit.



The mandate

Ascertain, with precision, precisely where orders erode capital — by platform, city, category, delivery interval, and discount — and distill this into judicious, actionable remedies.

A single corpus of data, three instruments — each deployed to its particular strength



I · FOUNDATION

SQL / PostgreSQL

Schema architecture and preliminary inquiry: revenue, average order value, principal cities and categories, payment composition.



II · DIAGNOSIS

Python / Pandas

Rigorous validation, feature derivation, segmental comparison, uplift analysis, and formal hypothesis testing.



III · EXPOSITION

Power BI

An interactive Margin Console — filterable by platform, city, category, distance, and rating.

947,752

TRANSACTIONS

21

ATTRIBUTES

8

PLATFORMS

12

CITIES

7

CATEGORIES

Scrutiny of the data returned an unimpeachable verdict



Absent values

Zero nulls across the entirety of 21 attributes



Duplication

Zero duplicate order identifiers ascertained



Aberrant values

Zero negative or null amounts, distances, or durations



Categorical fidelity

Uniform nomenclature across Platform, City, Category, Payment



The profit formula

Order Value less Delivery Cost less Discount — exact, without exception



Outlying values

?!1.2% of orders exceed the interquartile fence — retained, not excised

The singular figure against which every subsequent finding is adjudicated

♦ THE BENCHMARK

9.77%

OF ALL ORDERS INCUR ATTRITION

*Every segmental comparison in this exposition
is read against this benchmark — never in
isolation.*

♦ **Equipoise in sample composition**

Orders are distributed evenly across eight platforms (~118K apiece) and twelve cities (~7–8% each) — comparisons are commensurate, not skewed by scale.

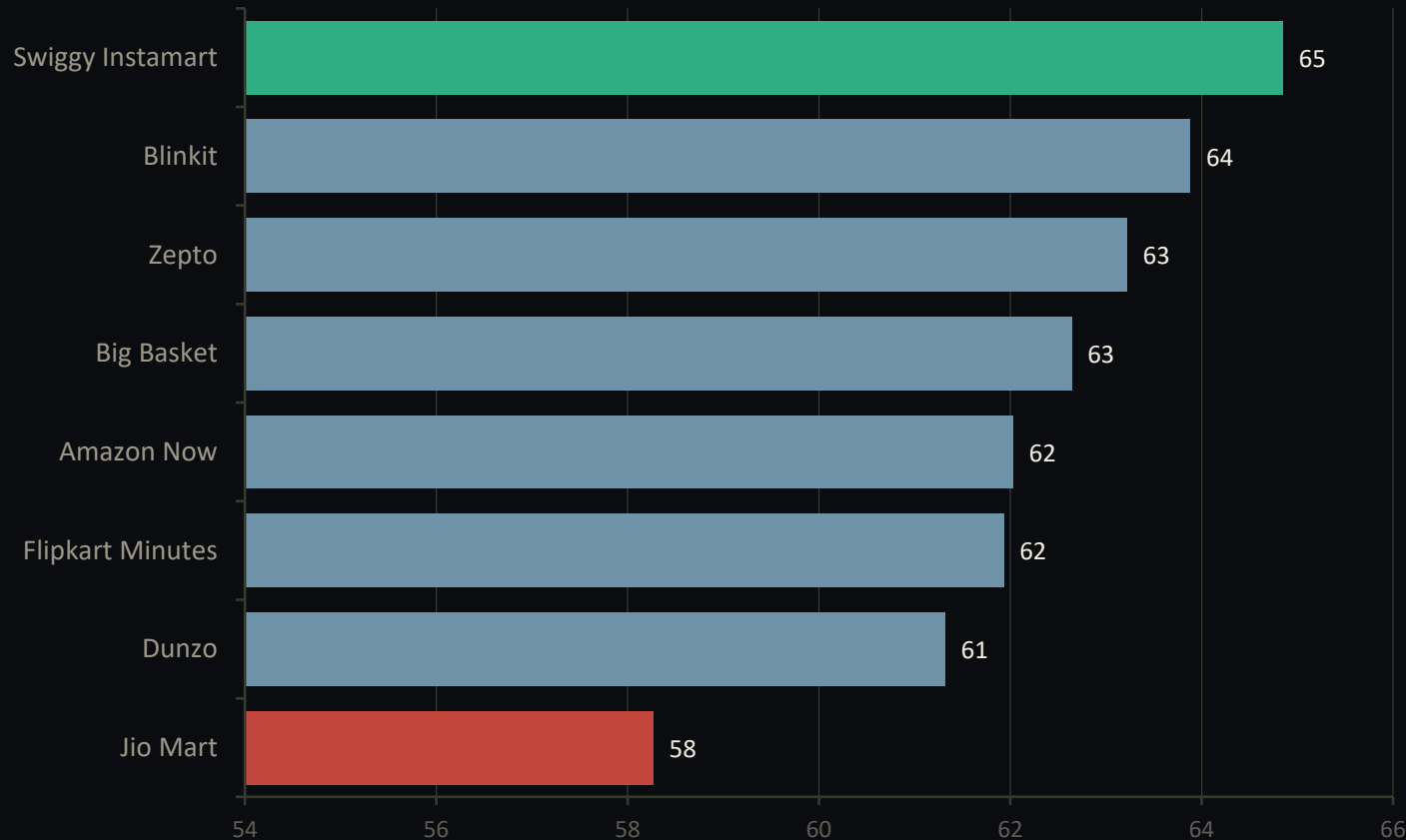
♦ **Order value governs profit**

A correlation coefficient of 1.00 between order value and profit — basket magnitude is the paramount determinant, eclipsing distance or delivery duration.

♦ **Distance: a modest, genuine drag**

A correlation of -0.09 between distance and profit — slight in isolation; it likely compounds with delivery cost and duration rather than acting unilaterally.

Jio Mart sustains the most attenuated margin of the eight platforms



6.6 PTS

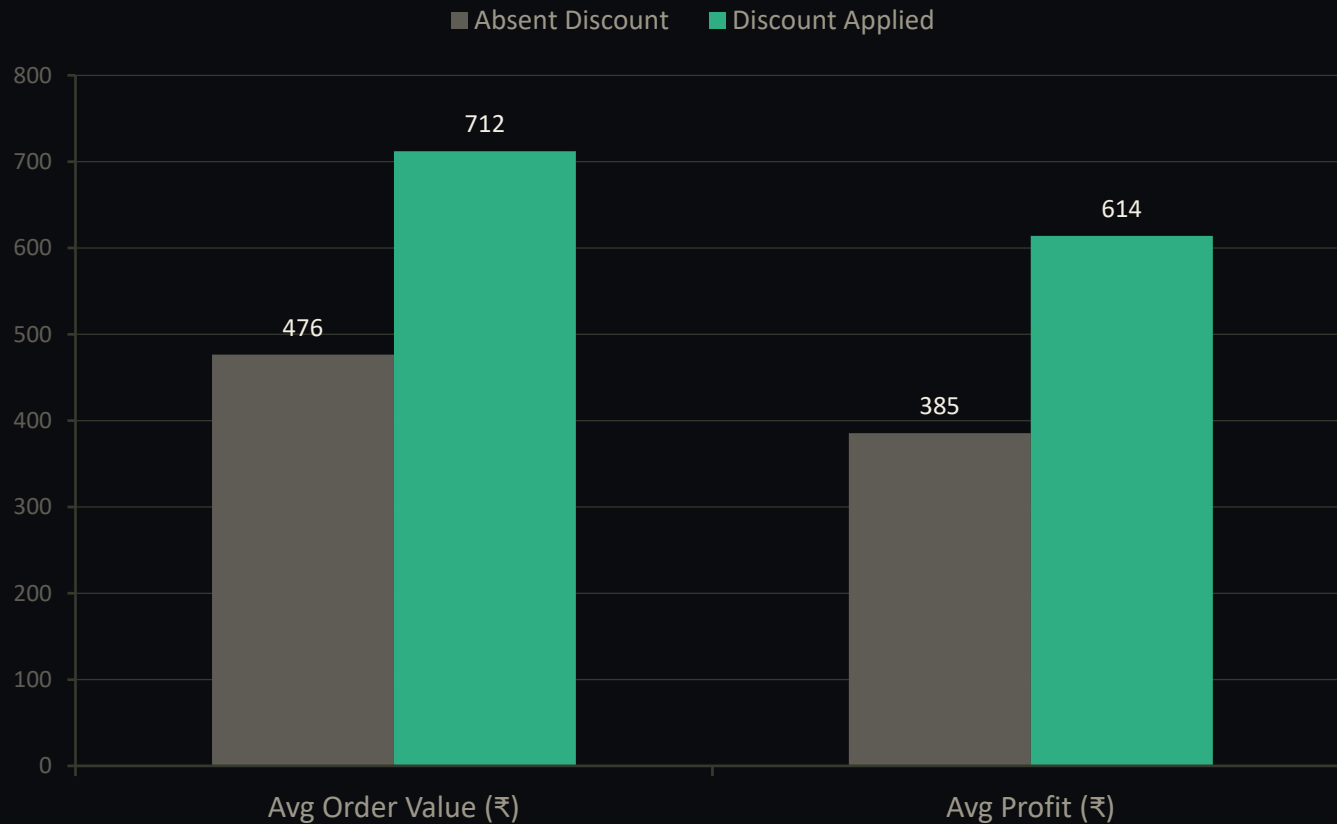
the disparity between Swiggy Instamart (64.9%, foremost) and Jio Mart (58.3%, hindmost)

Jio Mart likewise sustains the highest attrition rate at 10.81% of orders — exceeding the 9.77% benchmark — whilst Swiggy Instamart registers the lowest, at 9.21%.

► THE PRESCRIPTION

Audit Jio Mart's cost architecture — delivery expenditure, discount depth, basket composition — against Swiggy Instamart's, to narrow the disparity.

Discounted orders yield superior, not inferior, profit



+59.3%

superior average profit on discounted orders (₹614 versus ₹385)

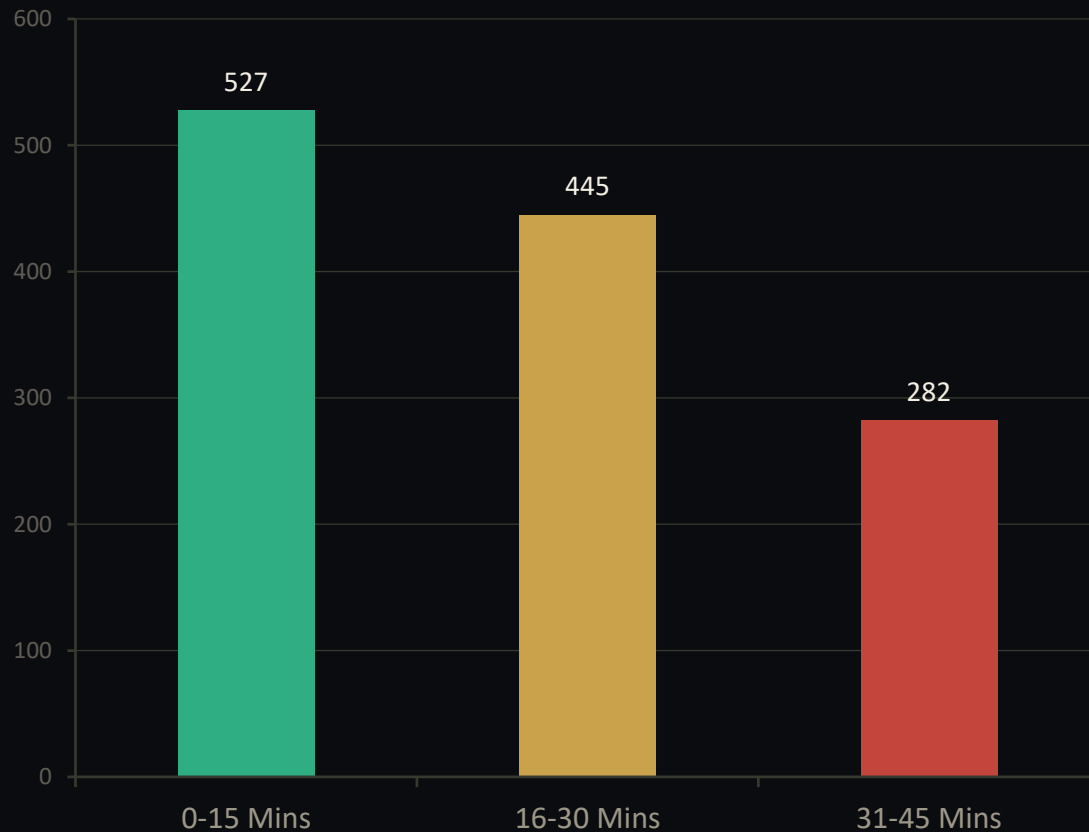
Corroborated with statistical rigour — Welch's t-test, $p < 0.001$, $n = 947,752$. A veritable effect, not statistical noise.

The effect is uniform across every age cohort (~49–50% uplift apiece) — age-based targeting will not differentiate outcomes.

► THE PRESCRIPTION

Retain the discounting strategy — pursue targeting by category or distance rather than age.

The swiftest orders yield 1.9x the profit — and incur attrition thrice less often



◆	Attrition, 0-15 min	6.88%
◆	Attrition, 16-30 min	11.54%
◆	Attrition, 31-45 min	23.07%

Delivery expenditure ascends with duration too — ₹76 on the swiftest tier versus ₹115 on the slowest — compounding the profit disparity atop the attrition disparity.

► THE PRESCRIPTION

Regard deliveries exceeding thirty minutes as a distinct operational exigency — examine distance, batching, or courier availability.

Every deficit-bearing order traces to a singular genesis

100%

OF DEFICIT-BEARING ORDERS ARE CLASSIFIED “LOW VALUE ORDER”

Distance and discounting remain correlated with diminished profit — yet the categorical Loss Driver attribute assigns every flagged deficit to basket magnitude alone.

92,592

DEFICIT ORDERS

-₹45.61

AVG DEFICIT/ORDER

₹4.22M

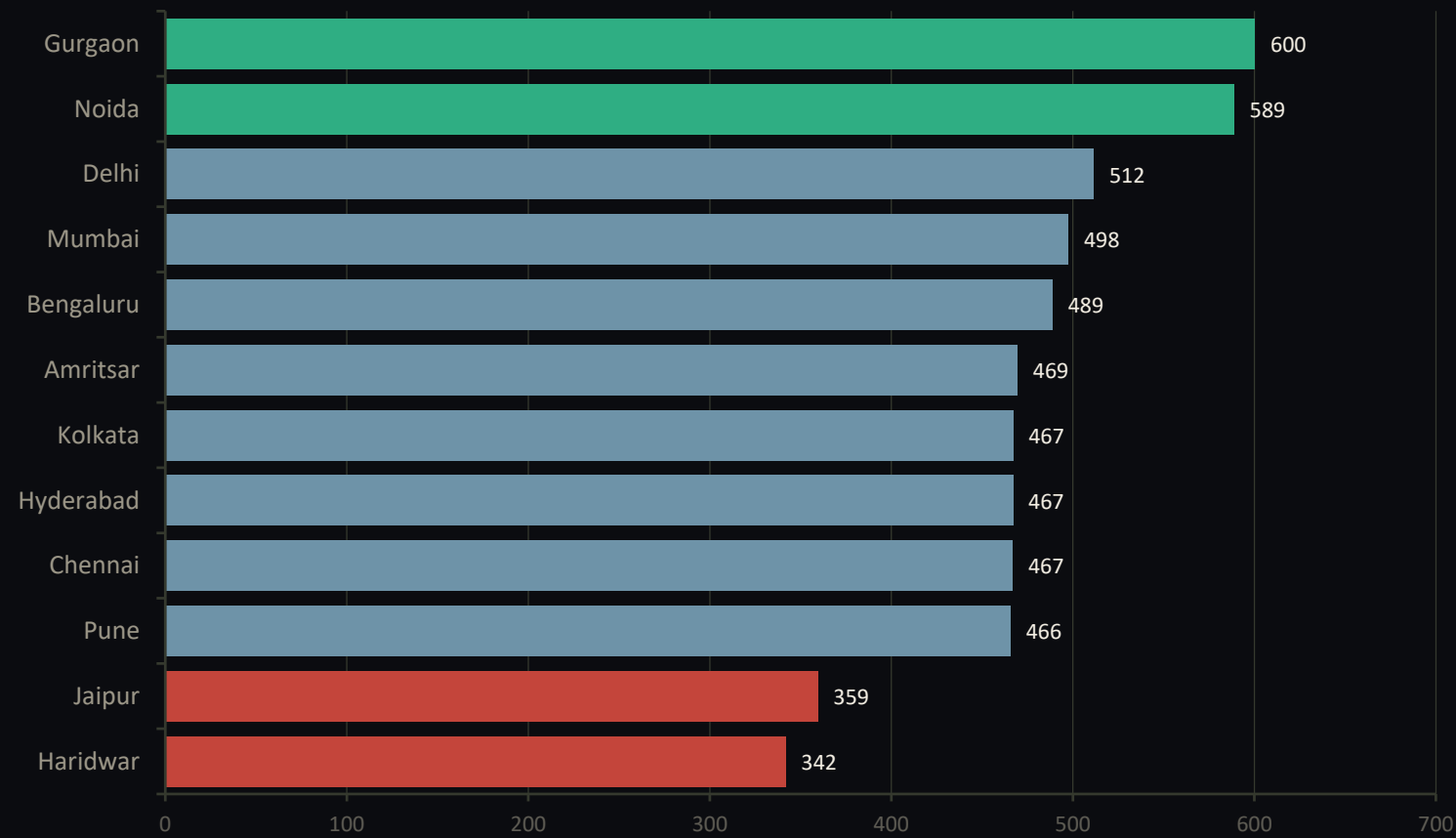
AGGREGATE DEFICIT

► THE PRESCRIPTION

Institute a minimum order value, or a delivery fee tiered to basket size — this singular measure addresses nearly the entirety of the identified deficit.

The most consequential remedy in the entire examination: a solitary lever, addressing approximately the whole of tracked attrition.

Haridwar and Jaipur falter across every platform alike



₹258 DISPARITY

between the foremost city (Gurgaon, ₹600) and the hindmost (Haridwar, ₹342) in average profit per order

This ranking persists uniformly across all eight platforms — no operator defies the civic trend, which precludes a platform-specific remedy.

► THE PRESCRIPTION

Regard Haridwar and Jaipur as market-level exigencies — examine local delivery cost architecture or demand density, not platform execution.

The principal findings withstand statistical scrutiny, not merely visual impression

THE INQUIRY	METHOD	RESULT	INTERPRETATION
◆ Discount versus absence thereof	Welch's t-test	t = 284.8, p < 0.001	Substantively significant — corroborates Revelation II
◆ Order Value ↔ Profit	Pearson correlation	r = 1.00	Basket magnitude is the paramount determinant
◆ Distance ↔ Profit	Pearson correlation	r = -0.09	Genuine, yet modest alone — compounds with cost & duration

► A MATTER OF CONSEQUENCE

With approximately 950,000 observations, virtually any disparity will appear “statistically significant.” We have paired each p-value with its effect size — the veritable rupee gap — so every prescription reflects genuine commercial consequence, not noise inflated by scale.

Five measures, ranked by their demonstrated commercial consequence

I



Remediate low-value orders foremost

Institute a minimum order value or a distance-tiered delivery fee — addresses ~100% of tracked attrition (₹4.22M, 92,592 orders).

II



Regard tardy deliveries as an operational exigency

31–45 min deliveries sustain a 23% attrition rate versus 6.9% for the swiftest tier — examine batching, staffing, courier availability.

III



Address Haridwar & Jaipur at the market level

Both cities falter across all eight platforms — examine local delivery cost or demand density, not platform execution.

IV



Audit Jio Mart's cost architecture

A 6.6-point disparity against Swiggy Instamart, plus the highest attrition rate (10.81%) — benchmark against the foremost performer.

V



Retain discounting; refine its targeting

Discounts elevate profit by 59%, with considerable statistical confidence — pursue targeting by category or distance, not age.

◆ THE EXAMINATION CONCLUDES

FROM 947,000 ENTRIES TO *FIVE JUDICIOUS REMEDIES*

The instrumentation — SQL, then Python, then Power BI — first affirmed the data's integrity, then advanced beyond dashboard-level averages to isolate *the genesis of each deficit* and corroborate it statistically:

- ◆ **9.77%** THE BENCHMARK ATTRITION RATE
 - ◆ **100%** OF ATTRITION TRACES TO LOW-VALUE ORDERS
 - ◆ **$p < 0.001$** DISCOUNTING CORROBORATED AS PROFIT-POSITIVE
 - ◆ **MARKET-WIDE** THE CIVIC EFFECT, NOT PLATFORM-SPECIFIC
-

HENCEFORTH: institute the minimum-order-value remedy, and reassess attrition next quarter.



Filter By

City: All

Amritsar	Delhi
Bengluru	Gurgaon
Chennai	Haridwar

Company: All

Amazon Now	Dunzo
Big Basket	Flipkart Minutes
Blinkit	Jio Mart

Product_Category: All

- Select all
- Beverages
- Dairy
- Fruits & Vegetables
- Groceries
- Household
- Personal Care
- Snacks

Distance_Km

0.5015.00

Customer Rating

15

Quick-Commerce Profitability · PostgreSQL → Excel → Power BI

Margin Console: Diagnosing Profit Leaks in Quick-Commerce

Where quick-commerce orders bleed money — replicated from a 1M+ row transactional dataset in the style of Swiggy Instamart, Blinkit, Zepto & BigBasket profitability reviews.

TOTAL REVENUE

₹541M

TOTAL PROFIT

₹427.8M

NET LOSS VALUE

₹4.40M

AVG PROFIT ORDER

₹451.37

AVG DELIVERY
COST/ORDER

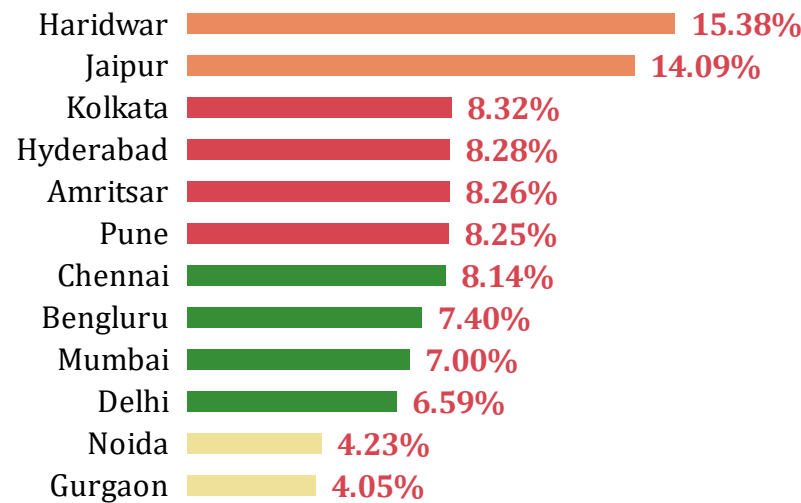
₹91.00

Discount Penetration %

40.08%

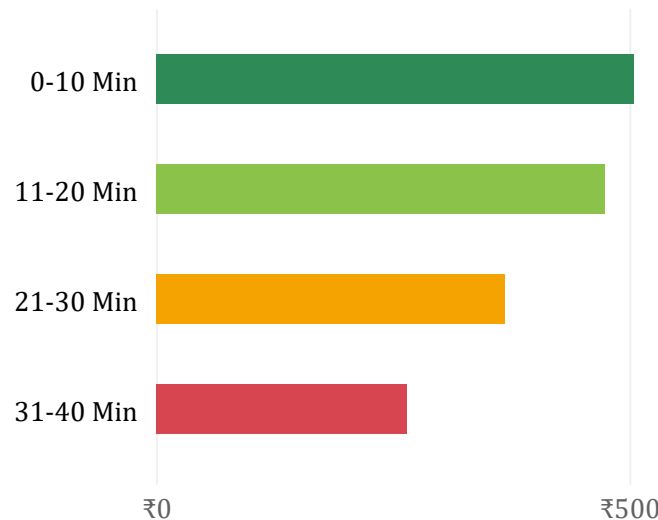
Loss leaderboard- by City

Top 3 cities drive 41.37% of total losses. Classic Pareto c...



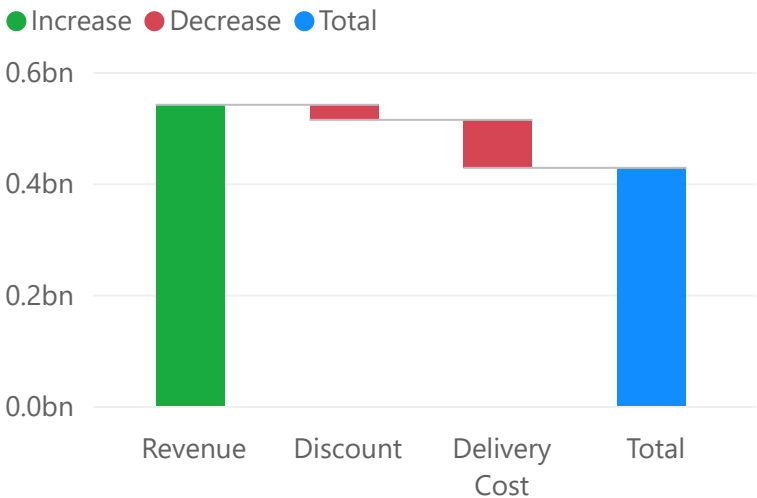
Average Profit by Delivery Time

Fastest orders ~2x more profitable than slowest.



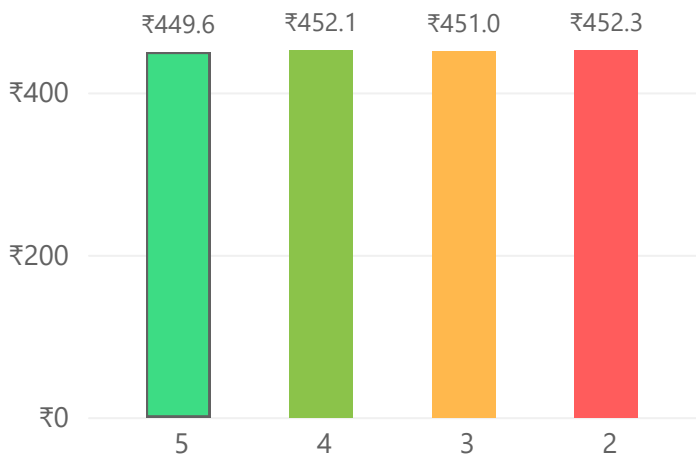
Net Profit

Discounts and delivery costs shave ~21% off revenue



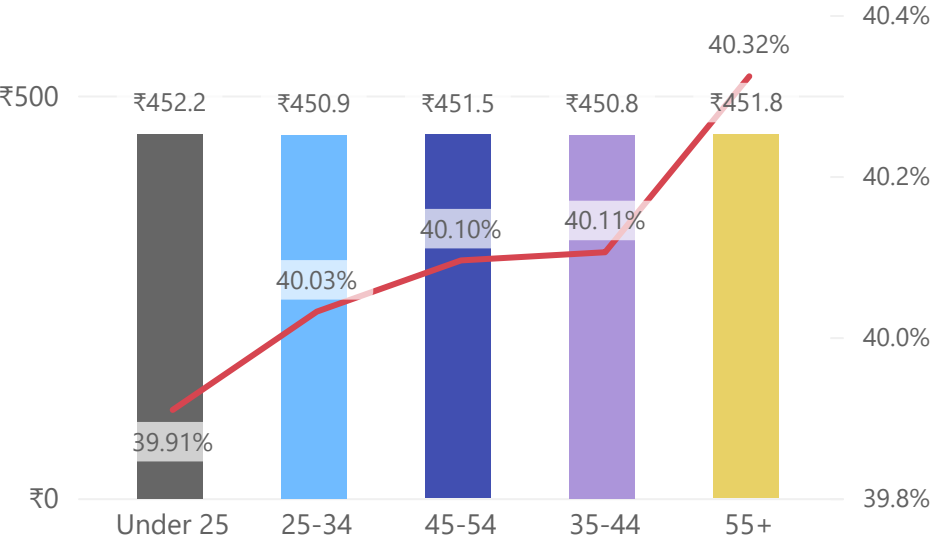
Average Profit by Delivery Partner Rating

Lower-rated partners correlate with lower profit...



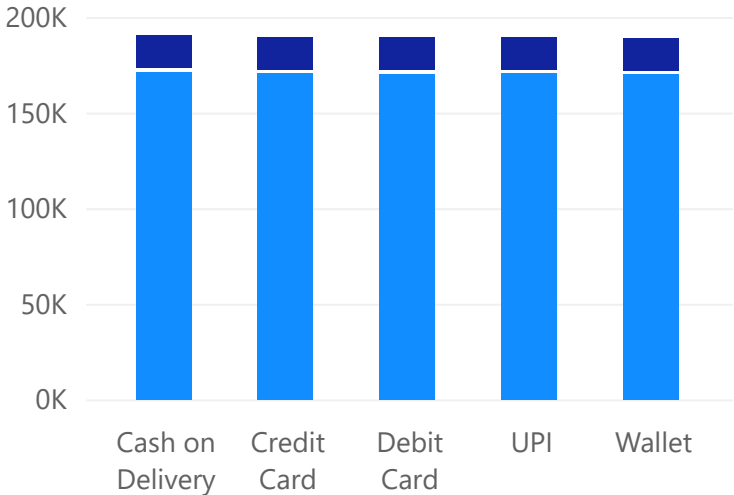
Profit vs Discount Usage by Age Group

Average of Profit Discount Usage %



Count of Order_ID by Payment_Method and Is_Loss

0 1



Category Loss and Cumulative Category Loss % by Product_Category

Losses are evenly spread — no single category dominates.

Category Loss Cumulative Category Loss %

